

Homework 2

- 1 Find the coordinate transformations between Poincare AdS_2 and global AdS_2 .
- 2 Write down the independent Killing vectors in global AdS_2 with the normalization

$$[L_1, L_{-1}] = 2L_0, \quad [L_0, L_{\pm 1}] = \mp L_{\pm 1} \quad (1)$$

- 3 Similarly, write down the independent Killing vectors in Poincare coordinate, denoted by $H_0, H_{\pm 1}$. Work out the relation between $\{H_i\}$ and $\{L_i\}$.
- 4 Draw Penrose diagram for global and Poincare AdS_2 .